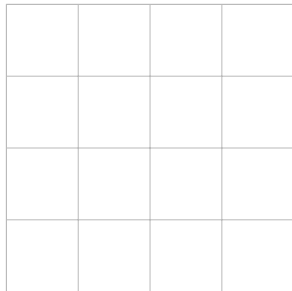
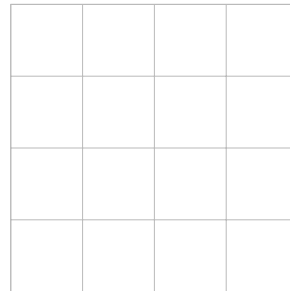


Keep all tiles flat and in one piece, joined along whole sides without overlaps. Count every side with no tile beside it, including sides around holes. Turns and flips count as the same shape.

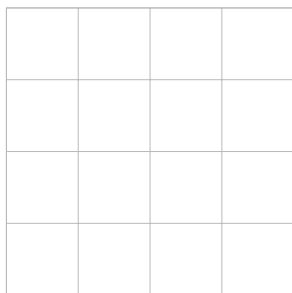
**Problem 1:** Make every different shape you can with four tiles. Draw each shape and count its boundary sides.



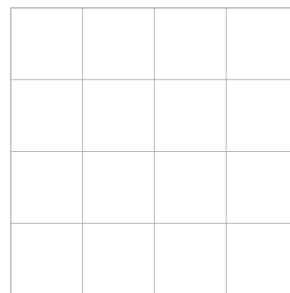
\_\_\_\_\_ sides



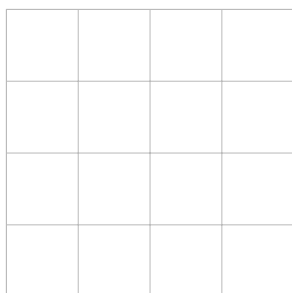
\_\_\_\_\_ sides



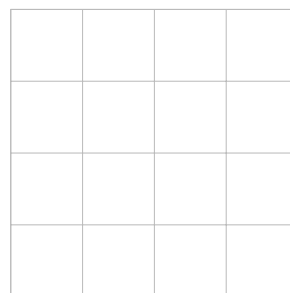
\_\_\_\_\_ sides



\_\_\_\_\_ sides

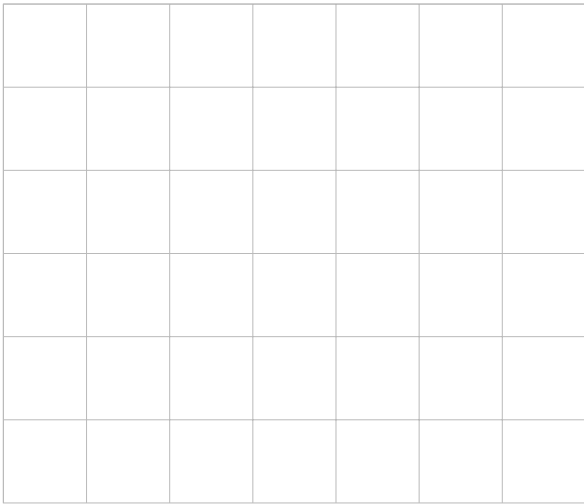


\_\_\_\_\_ sides

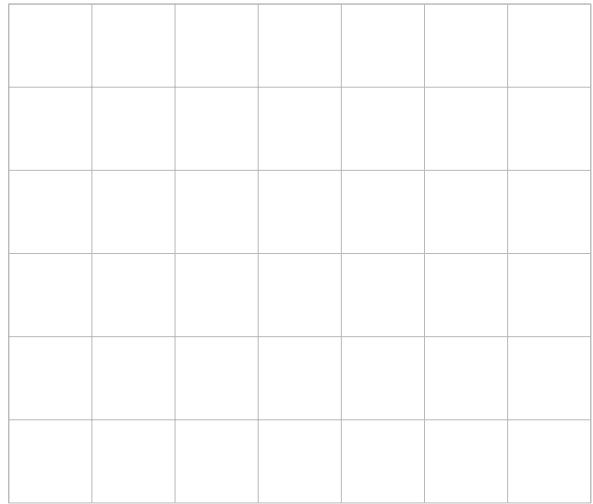


\_\_\_\_\_ sides

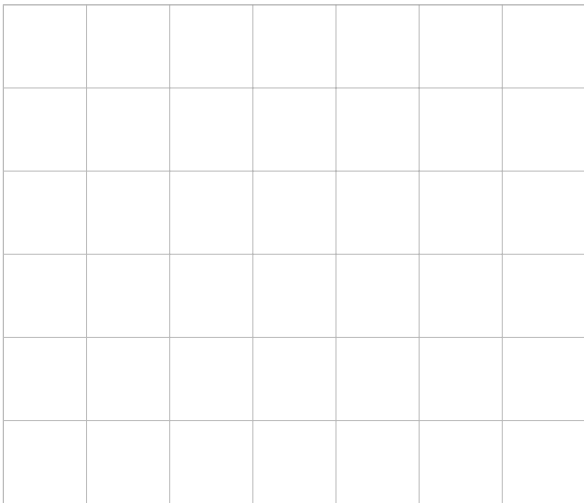
**Problem 2:** Make a shortest-boundary shape and a longest-boundary shape with 5 tiles. Do the same with 6 tiles.



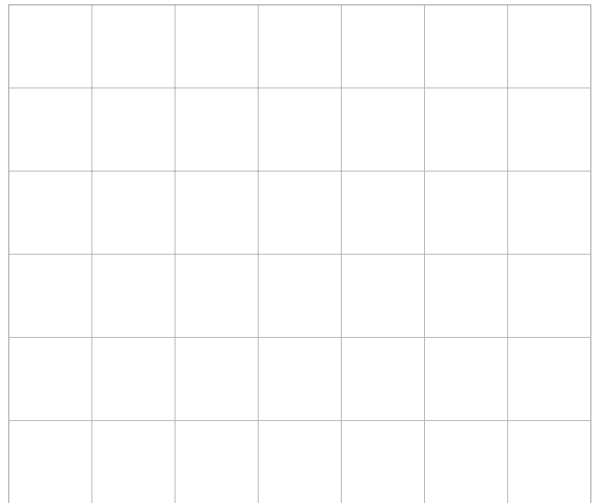
5 tiles; shortest  
\_\_\_\_\_ sides



5 tiles; longest  
\_\_\_\_\_ sides



6 tiles; shortest  
\_\_\_\_\_ sides

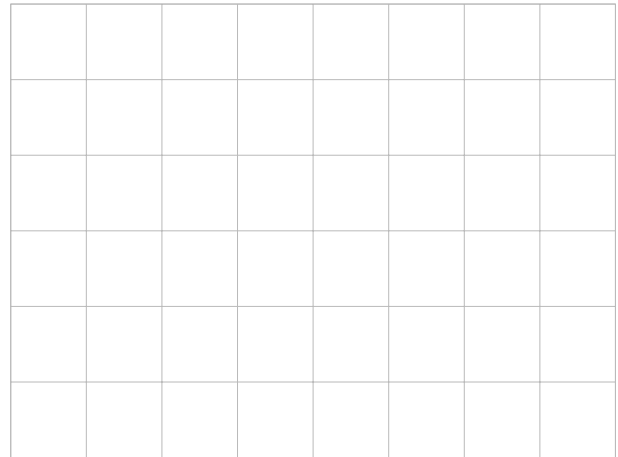


6 tiles; longest  
\_\_\_\_\_ sides

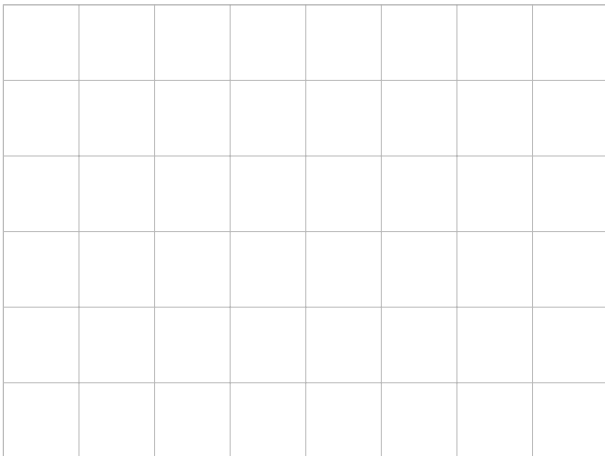
**Problem 3:** Make a shortest-boundary shape and a longest-boundary shape with 7 tiles. Do the same with 8 tiles.



7 tiles; shortest  
\_\_\_\_\_ sides



7 tiles; longest  
\_\_\_\_\_ sides

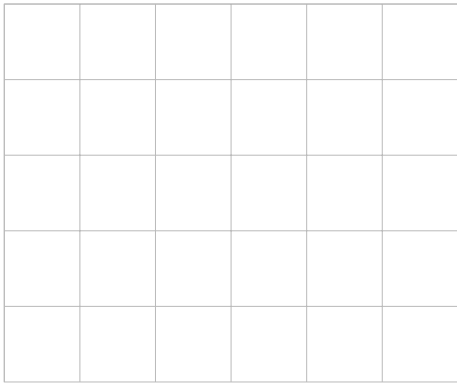


8 tiles; shortest  
\_\_\_\_\_ sides

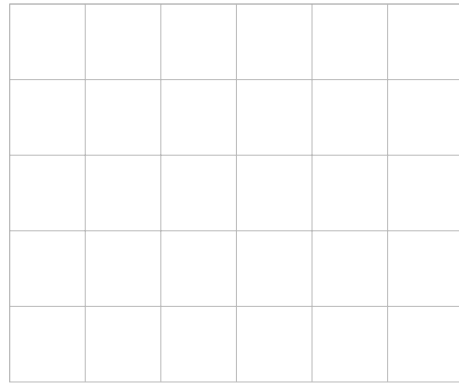


8 tiles; longest  
\_\_\_\_\_ sides

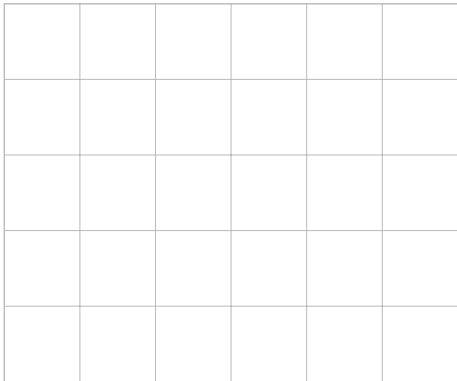
**Problem 4:** Use six tiles to make shapes with 10, 12, and 14 boundary sides. Which numbers can you make in two different ways?



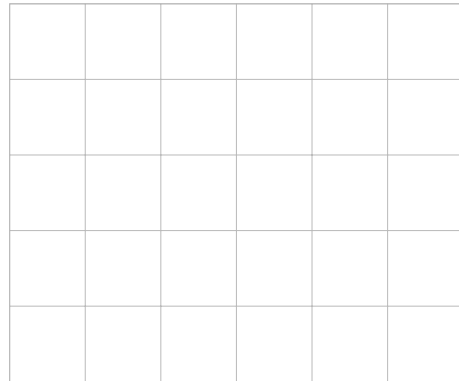
10 sides



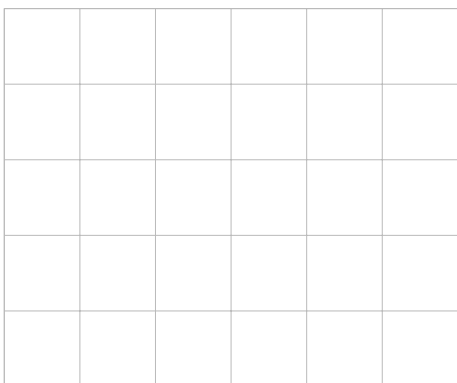
10 sides



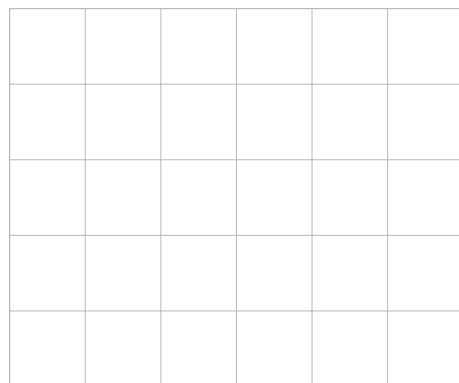
12 sides



12 sides

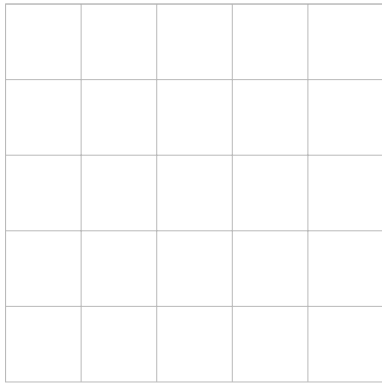


14 sides

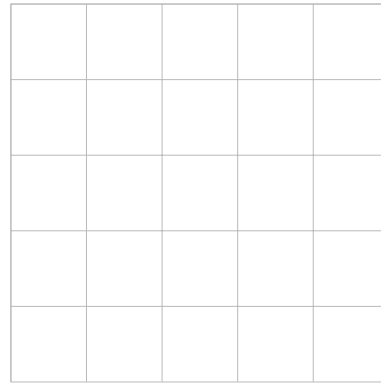


14 sides

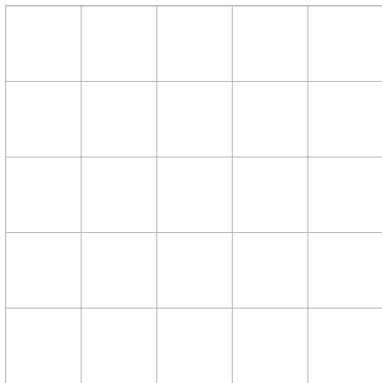
**Problem 5:** Which of these boundary lengths can you make with five tiles? Make each one you can, and tell your adult why the others will not work.



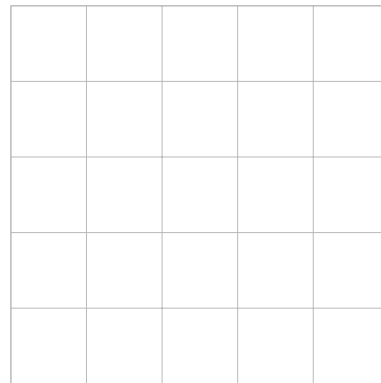
8 sides



9 sides



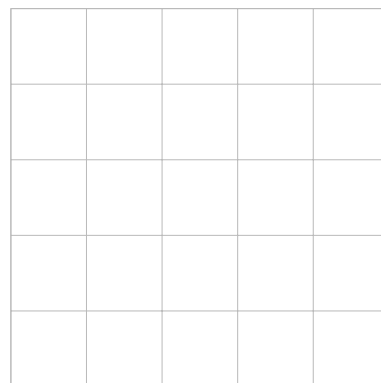
10 sides



11 sides

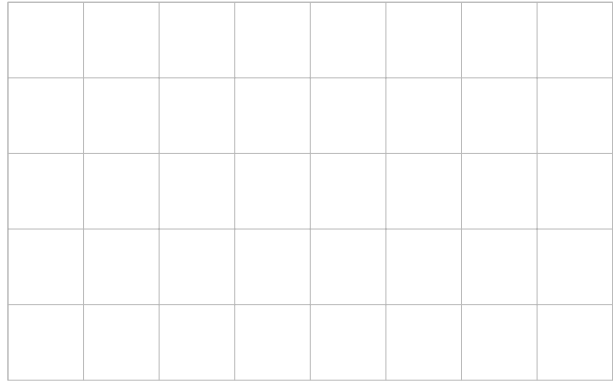
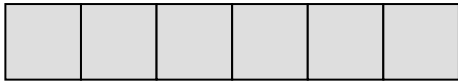


12 sides

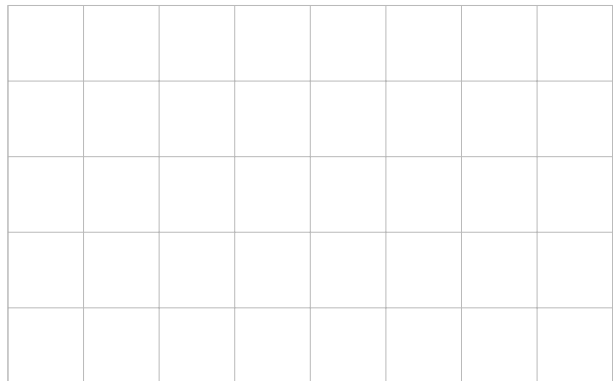
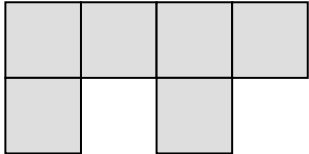


13 sides

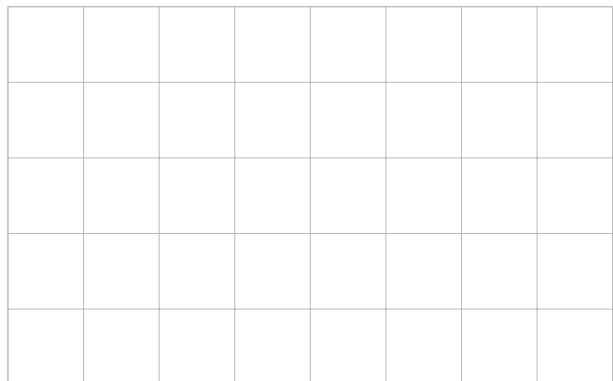
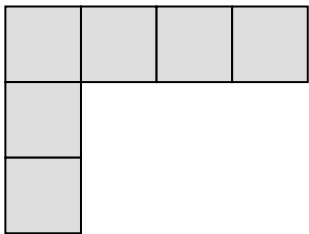
**Problem 6:** Move one tile in each shape and draw a result with the shortest boundary you can make. Which shapes can you make shorter?



\_\_\_\_\_ sides



\_\_\_\_\_ sides



\_\_\_\_\_ sides